REFERENCE GUIDE · KEEP WITH YOU ON THE JOB

Data Hall 4 Cable Installation

Large site, data hall 4: pods 10, 11, 12 · 144F and 192F trunks · network rows 1-2.

Built from the Large Site Cable Installation Guide (D. Moore; cover Rev D, page footers Rev C, 10/8/2026) and the Fluidstack DRS sheets added so far (DRS 01.00.00 to 01.00.03.01). Companion to the interactive beginner course. A study aid: never a substitute for the DRS.

The current DRS says how the work must be done and must be on hand where you work. Your site's cutsheet is the final word on any port. If a label, this guide, the DRS or the cutsheet disagree, stop and ask your lead.

Fluidstack Internal Use Only content. Keep within the project.

Contents: DRS sheets Connection tables Done-when checklists Open items Glossary Sources & coverage

DRS sheets in this course

Every Fluidstack DRS sheet added so far, in plain words. The current DRS from the portal is the requirement: check its revision date and have it on hand.

`DRS 01.00.00` Introduction: what a DRS is

Portal page · approved 6/9/2026 · approved by Tom Marsh, Infrastructure Deployment Manager

1. DRS = Deployment Requirement Sheet: Fluidstack's technical standards for how installation work must be done. **Requirements, not guidelines.** Every on-site task has a matching DRS.
2. Contains step-by-step requirements, specifications, approved materials, diagrams, and references to ANSI/TIA, BICSI, NEC and NEMA standards.
3. Exists so every deployment meets the same quality standard, whoever does the work; protects equipment and lowers outage risk.
4. Approved by Fluidstack ICT leadership. New revisions are announced through the portal. **Always use the current version: check the revision date on the cover page.**
5. Get it from Fluidstack Confluence (fluidstack.atlassian.net → LVC Deployment space → DEPLOYMENT REQUIREMENTS SHEETS). Backup: your Fluidstack PM or site supervisor.
6. **On site:** have the applicable DRS on hand, printed or on a device. Working from memory alone is not acceptable. No DRS: **stop work and get it.**

Standards named: ANSI/TIA, BICSI, NEC, NEMA

Taught in: Lesson 0 · What a DRS is · Lesson 0 · Getting the right DRS · Lesson 0 · The on-site rule · **Video:** Lesson 0 video · **Open items:** 14, 17

`DRS 01.00.01` General Fiber Cable Dressing

Drawing + portal summary · portal title: General Cable Dressing · revised 6/9/2026 · approved by Tom Marsh

1. Dress cables in accordance with **ANSI/TIA-568, ANSI/TIA-606 and ANSI/TIA-942**.
2. Bundle with **1/2-inch hook-and-loop (Velcro)**. Nylon zip ties are not allowed.
3. **Bend radius:** follow the manufacturer's spec; minimum **20 × diameter loaded** (under tension), **10 × unloaded**. See DRS 01.00.07 for fiber bend radius.
4. Cable supports must not damage the cable: **no metal staples**.
5. **Support every 4-5 feet** on horizontal runs (TIA-568 recommendation).
6. **Rated cable for the space:** plenum OFNP / CMP in plenum spaces, riser OFNR / CMR in vertical risers, LSZH where local code requires.

Standards named: ANSI/TIA-568, ANSI/TIA-606, ANSI/TIA-942, NEC (cable ratings)

Taught in: Lesson 5 · The six rules · Lesson 5 · The standards behind the rules · Lesson 5 · Ties, bends and supports · Lesson 5 · Rated cable · Lesson 5 · Bend radius worked example · Lesson 3 · Handling fiber · **Video:** Lesson 5 video; Lesson 3 video (Handling fiber) · **Open items:** 12, 13, 15, 16

`DRS 01.00.02` Front of Rack Cable Dressing

Drawing + portal summary · revised 6/9/2026 · approved by Tom Marsh

1. Cables shall **not cross devices they do not patch into**. Every device must be replaceable without unpatching cables that end in another device.
2. Cable shall **exit the basket tray directly above the rack edge or the vertical cable managers** (VCMs).
3. All cables entering or leaving the basket tray must be **supported by waterfalls** to keep the proper bend radius.
4. **Over-the-top waterfalls** must be used for **network racks**.
5. Bundle cable neatly with **1/2-inch Velcro**.
6. Cables should use the **waterfall next to the side of the rack they patch into**.
7. Drawing note: each waterfall lines up with the rack VCM. **Tray accessories must fit** trays made by Basor, WBT, Legrand or equivalent, with 2 × 4 inch (50 × 100 mm) openings and 5 mm wire.

Taught in: Lesson 5B · Front of the rack · Lesson 5B · Worked example · Lesson 9 · Landing the 192F (network rack) · Lesson 9 · Cords to the spines (network rack) · **Video:** Lesson 5B video (Front of the rack) · **Open items:** 12, 18, 19

`DRS 01.00.03` Cable Dressing Basket Trays

Drawing + portal summary · portal title: Basket Tray Cable Dressing · revised 6/9/2026 · approved by Tom Marsh

1. Bundle cables neatly in the tray with **Velcro 12 to 18 inches apart**.
2. **Do not secure bundles to the basket tray** except going around bends, exiting the tray, or running vertically. **Secure 3× at corners and 3× at exits.**
3. **1/2-inch Velcro** for all cable bundling.
4. **Waterfalls centered over the rack VCMs.**
5. Cables keep the **manufacturer's recommended bend radius** at all times. Drawing: fiber bend radius at a tray corner shown as **20 × cable diameter**.
6. Portal summary: **never allow cable to drop over a tray edge unsupported**. Waterfalls keep the bend radius at every tray exit.

Taught in: Lesson 5B · From the tray to the device · Lesson 5B · In the basket tray · Lesson 5B · Worked example · Lesson 11 · Pulling and dressing the OCS bundles · Lesson 12 · Routing and labels · **Video:** Lesson 5B video (In the basket tray); Lesson 11 video (Pulling and dressing) · **Open items:** 12, 20

`DRS 01.00.03.01` Cable Dressing Vertical Pathways

Drawing + portal summary · portal title: Vertical Pathway Dressing · revised 6/9/2026 · approved by Tom Marsh

1. Bundle cables neatly with **Velcro 12 to 18 inches apart**, in vertical tray as well.
2. **Secure bundles to the cable tray when running vertically** (horizontal bundles float).
3. Cable in vertical tray or ladder must be **secured to be self-supporting**. Excessive cable weight on horizontal transitions damages the tray and the cable.
4. **Waterfalls are required at all vertical transitions** to keep the cable bend radius. A missing waterfall can damage the cable.
5. **3/4-inch Velcro may be used to secure cable to vertical tray.**
6. Drawing: **all bends greater than 20 × the cable diameter**. Portal: **too little Velcro on vertical runs lets cable drop to the bottom** under its own weight, damaging cable and tray.

Taught in: Lesson 5B · From the tray to the device · Lesson 5B · Vertical pathways · Lesson 5B · Worked example · Lesson 5 · Velcro sizes · Lesson 11 · Pulling and dressing the OCS bundles · **Video:** Lesson 5B video (Vertical pathways) · **Open items:** 12, 20

Referenced, not yet in the course:

- DRS 01.00.07: Fiber bend radius. DRS 01.00.01 says "see DRS 01.00.07".
- DRS 01.00, 01.01, 02.00, 05.00, 10.01, 10.04, 10.08, 10.10: The DRS sections the installation guide says it was built from (guide p.1). Which 3-part sheets belong to each is not stated.

Connection tables and rules

The current DRS says how the work must be done; your site's cutsheet is the final word on any port. These tables come from the guide (cover Rev D, footers Rev C, 10/8/2026) and the DRS sheets added so far (01.00.00 to 01.00.03.01). They are a study aid, not a substitute for the DRS. If anything here disagrees with the current DRS or your cutsheet, follow the DRS and the cutsheet and tell your lead.

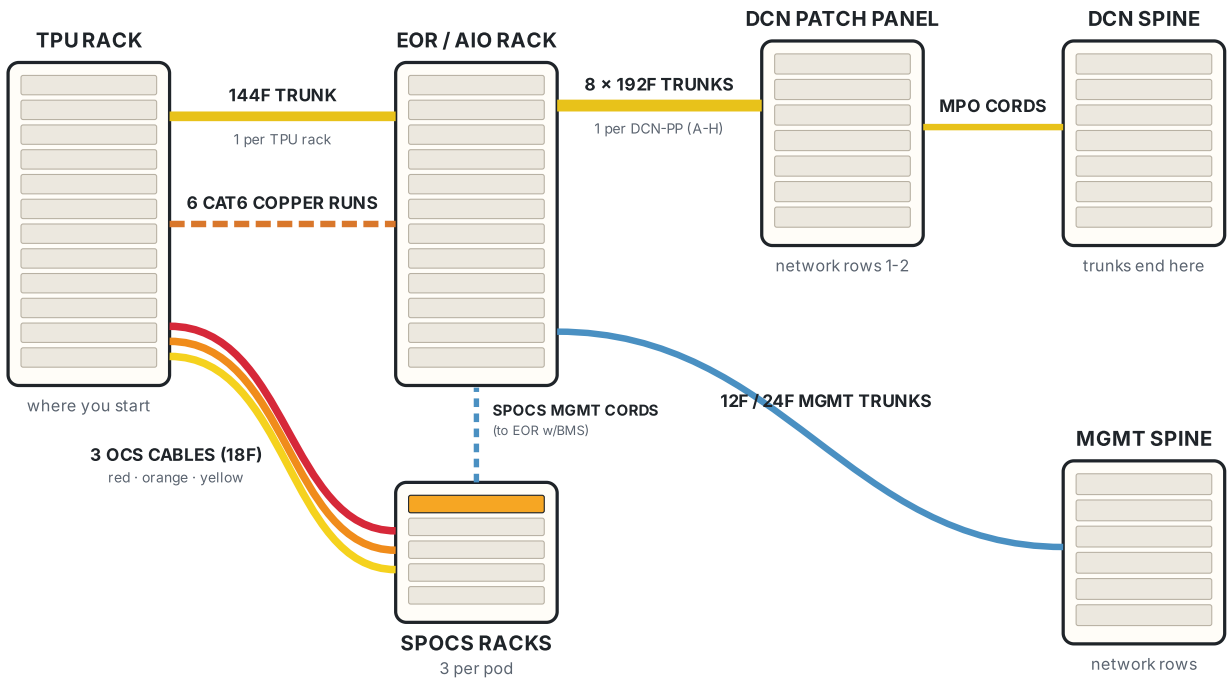
0. The DRS: Deployment Requirement Sheets (DRS 01.00.00)

- **What:** Fluidstack's technical standards for how installation work must be done. **Requirements, not guidelines.** Every on-site task has a matching DRS.
- **Contains:** step-by-step requirements, specifications, approved materials, diagrams, and references to ANSI/TIA, BICSI, NEC and NEMA standards.
- **Approved by** Fluidstack ICT leadership. **Revisions** are announced through the portal: always use the current version and check the revision date on the cover page.
- **Where:** Fluidstack Confluence, `fluidstack.atlassian.net` → LVC Deployment space → DEPLOYMENT REQUIREMENTS SHEETS. **Backup:** your Fluidstack PM or site supervisor (they can send the document or restore access).
- **On site:** the applicable DRS must be on hand where you work, printed or on a device. **Working from memory alone is not acceptable. No DRS: stop work and get it.**
- **Internal Use Only:** DRS documents are confidential Fluidstack property. Do not copy, post or share them outside the job.
- **Sheets in this course so far:** 01.00.00 (introduction), 01.00.01 (general cable dressing), 01.00.02 (front of rack cable dressing), 01.00.03 (basket tray cable dressing), 01.00.03.01 (vertical pathways). See the DRS sheets list.

Document	Answers
DRS (current revision)	How each task must be done

Document	Answers
Cutsheet	Where each cable goes: the final word on any port
Installation guide	Built from the DRS and the BUF102 matrix v0.97; explains the system
This course, reference, Field Kit	Training and checking aids only

1. The five cable paths (guide p.4)



The five cable paths your crew installs.

#	Cable	From → to	Lessons
1	144F trunk (1 per TPU rack)	TPU rack TOR odd ports 1-31 → EOR/AIO MPO panel front (RU 32)	7, 8

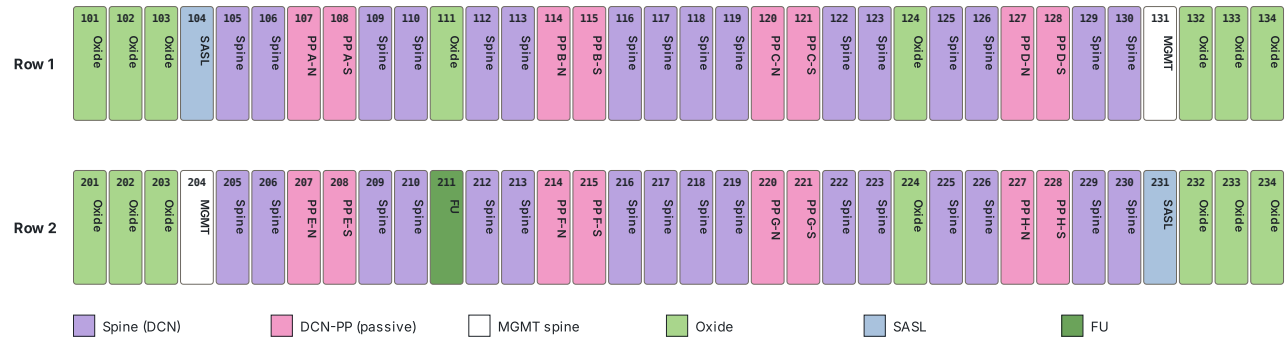
#	Cable	From → to	Lessons
2	8 × 192F trunks per AIO + MPO cords	AIO MPO panel rear → DCN-PP racks (rows 1-2) → DCN spines	8, 9
3	3 × 18F OCS cables per TPU rack	pp1 ports 1-8 / 9-16 / 17-24 → SPOCS racks 1 / 2 / 3 of the pod	10, 11
4	6 × CAT6 home runs per TPU rack	TOR ETH + CON, RMU1 and RMU2 GIG E 1 + CON → EOR copper panels RU 24 / 30 / 22	12
5	12F / 24F mgmt trunks + SPOCS mgmt cords	AIO RU 39 LC panel → management spines R131 / R204; SPOCS mgmt1/mgmt2 → w/BMS EOR	13, 14

Do not touch: anything inside the TPU rack (machines, trays, DAC and CDFP cables, batteries), or anything past the spines.

2. Hall map facts (guide p.5, 18, 25, 33, 44)

Location code	SITE-DH4-R0512 = site · hall 4 · row 05 · spot 12. Map shows row+spot: 512. Last two digits = spot.
Rows	Network rows 1-2 (west end). Compute rows 3-26. Rows pair across hot aisles; even rows mirrored.
Spots in every compute row	02-10: 9 TPU racks (north) · 11: EOR for north · 12: middle aisle (SPOCS in some rows) · 13: EOR for south · 14-22: 9 TPU racks (south). CDU blocks at row ends (not yours).
Which EOR?	Spots 2-10 → spot 11 of the same row. Spots 14-22 → spot 13.
Pods	144 TPU racks sharing 3 SPOCS racks. Pod 10 rows 3-10 · pod 11 rows 11-18 · pod 12 rows 19-26. 432 TPU racks, 48 AIO racks in the hall.
SPOCS racks (spot 12)	Pod 10: 512 red · 612 orange · 712 yellow. Pod 11: 1312 · 1412 · 1512. Pod 12: 2112 · 2212 · 2312.

EOR w/BMS racks	611, 613, 1411, 1413, 2211, 2213 (24F management trunks, FB0 BMS switch).
DCN-PP pairs (N / S)	A 107/108 · B 114/115 · C 120/121 · D 127/128 · E 207/208 · F 214/215 · G 220/221 · H 227/228.
Management spines	R131 and R204.



Network rows 1 and 2.

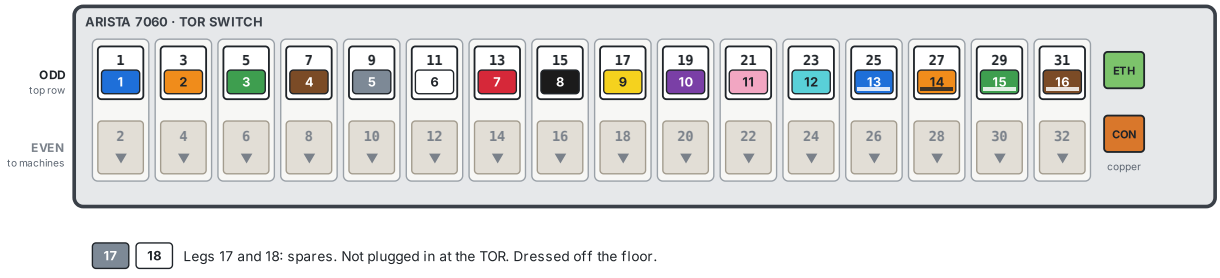
3. 144F trunk at the TOR (guide p.13) and AIO front (p.36)

TOR port = 2 × leg − 1 · legs 17-18 spare

Leg	Color	TOR port	AIO front column
1	Blue	**1**	A (port 1)
2	Orange	**3**	A (port 2)
3	Green	**5**	B (port 3)
4	Brown	**7**	B (port 4)
5	Slate	**9**	C (port 5)

Leg	Color	TOR port	AIO front column
6	White	**11**	C (port 6)
7	Red	**13**	D (port 7)
8	Black	**15**	D (port 8)
9	Yellow	**17**	E (port 9)
10	Violet	**19**	E (port 10)
11	Rose	**21**	F (port 11)
12	Aqua	**23**	F (port 12)
13	Blue (2nd round)	**25**	G (port 13)
14	Orange (2nd round)	**27**	G (port 14)
15	Green (2nd round)	**29**	H (port 15)
16	Brown (2nd round)	**31**	H (port 16)
17	Slate (2nd round)	**spare — not plugged**	I (port 17)
18	White (2nd round)	**spare — not plugged**	I (port 18)

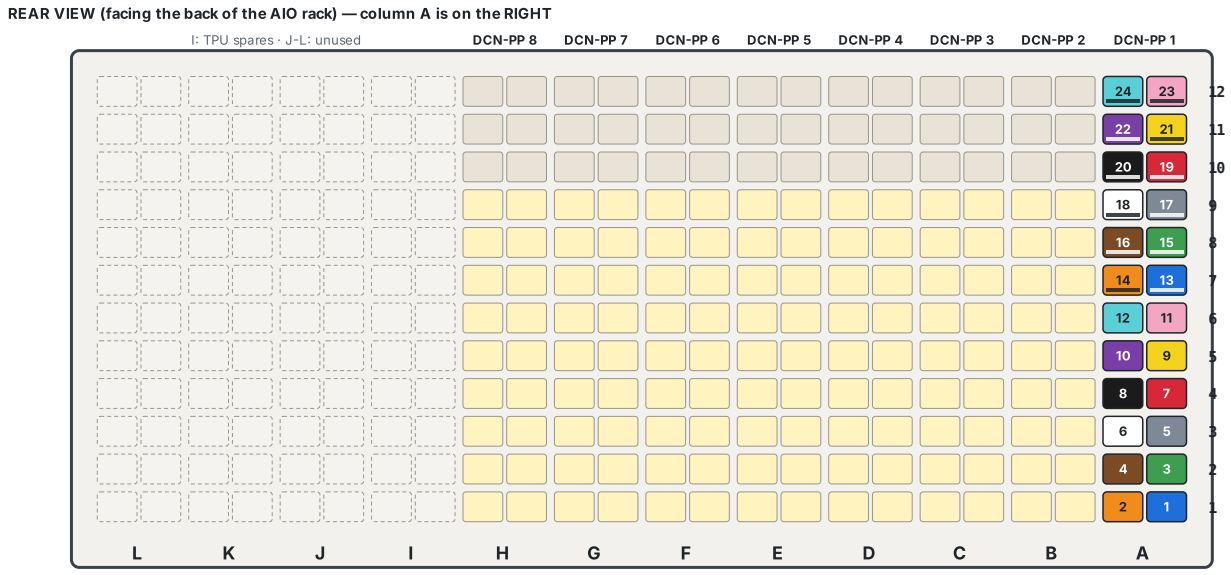
AIO front: **one TPU rack per row** (lowest-numbered rack on row 1 … row 9), legs left to right, legs 17-18 in column I, rows 10-12 and columns J-L open. Label format `(column)(port)-(row)`: TPU 3 leg 5 = `C5-3`.



TOR switch with all 16 legs in the odd ports.

4. 192F trunks: AIO rear (guide p.37)

- Facing the rear, column A is on the right. One 192F per column: **A = DCN-PP 1 (A) … H = DCN-PP 8 (H)**. Column I = TPU spares; J-L open.
- Legs go up: legs 1-2 row 1 … 23-24 row 12. Odd legs on the right port, even on the left (rear view). Legs 19-24 (rows 10-12) spares.
- Front row N meets rear row N: TPU N's legs 1-2 → DCN-PP 1 (trunk legs 2N−1, 2N) … legs 15-16 → DCN-PP 8.



AIO MPO panel, rear view.

5. Trunk Landing Sheet — every AIO rack (guide p.44-45)

Spot 11 AIO → N rack; spot 13 → S rack. Rows 3-14 → RU 35 panel; rows 15-26 → RU 39 panel (halls 1-3 land in RU 11-31). Lane = AIO-PP number · column pair · lane 1 or 2.

AIO rack	DCN-PP panel	Lane	Col A	B	C	D	E	F	G	H
R0311	RU 35	AIO-PP1 · AB #1	R107	R114	R120	R127	R207	R214	R220	R227
R0411	RU 35	AIO-PP2 · AB #2	R107	R114	R120	R127	R207	R214	R220	R227
R0511	RU 35	AIO-PP3 · CD #1	R107	R114	R120	R127	R207	R214	R220	R227
R0611 bms	RU 35	AIO-PP4 · CD #2	R107	R114	R120	R127	R207	R214	R220	R227
R0711	RU 35	AIO-PP5 · EF #1	R107	R114	R120	R127	R207	R214	R220	R227
R0811	RU 35	AIO-PP6 · EF #2	R107	R114	R120	R127	R207	R214	R220	R227
R0911	RU 35	AIO-PP7 · GH #1	R107	R114	R120	R127	R207	R214	R220	R227
R1011	RU 35	AIO-PP8 · GH #2	R107	R114	R120	R127	R207	R214	R220	R227
R1111	RU 35	AIO-PP9 · IJ #1	R107	R114	R120	R127	R207	R214	R220	R227
R1211	RU 35	AIO-PP10 · IJ #2	R107	R114	R120	R127	R207	R214	R220	R227
R1311	RU 35	AIO-PP11 · KL #1	R107	R114	R120	R127	R207	R214	R220	R227
R1411 bms	RU 35	AIO-PP12 · KL #2	R107	R114	R120	R127	R207	R214	R220	R227
R1511	RU 39	AIO-PP1 · AB #1	R107	R114	R120	R127	R207	R214	R220	R227
R1611	RU 39	AIO-PP2 · AB #2	R107	R114	R120	R127	R207	R214	R220	R227

AIO rack	DCN-PP panel	Lane	Col A	B	C	D	E	F	G	H
R1711	RU 39	AIO-PP3 · CD #1	R107	R114	R120	R127	R207	R214	R220	R227
R1811	RU 39	AIO-PP4 · CD #2	R107	R114	R120	R127	R207	R214	R220	R227
R1911	RU 39	AIO-PP5 · EF #1	R107	R114	R120	R127	R207	R214	R220	R227
R2011	RU 39	AIO-PP6 · EF #2	R107	R114	R120	R127	R207	R214	R220	R227
R2111	RU 39	AIO-PP7 · GH #1	R107	R114	R120	R127	R207	R214	R220	R227
R2211 bms	RU 39	AIO-PP8 · GH #2	R107	R114	R120	R127	R207	R214	R220	R227
R2311	RU 39	AIO-PP9 · IJ #1	R107	R114	R120	R127	R207	R214	R220	R227
R2411	RU 39	AIO-PP10 · IJ #2	R107	R114	R120	R127	R207	R214	R220	R227
R2511	RU 39	AIO-PP11 · KL #1	R107	R114	R120	R127	R207	R214	R220	R227
R2611	RU 39	AIO-PP12 · KL #2	R107	R114	R120	R127	R207	R214	R220	R227
R0313	RU 35	AIO-PP1 · AB #1	R108	R115	R121	R128	R208	R215	R221	R228
R0413	RU 35	AIO-PP2 · AB #2	R108	R115	R121	R128	R208	R215	R221	R228
R0513	RU 35	AIO-PP3 · CD #1	R108	R115	R121	R128	R208	R215	R221	R228
R0613 bms	RU 35	AIO-PP4 · CD #2	R108	R115	R121	R128	R208	R215	R221	R228
R0713	RU 35	AIO-PP5 · EF #1	R108	R115	R121	R128	R208	R215	R221	R228

AIO rack	DCN-PP panel	Lane	Col A	B	C	D	E	F	G	H
R0813	RU 35	AIO-PP6 · EF #2	R108	R115	R121	R128	R208	R215	R221	R228
R0913	RU 35	AIO-PP7 · GH #1	R108	R115	R121	R128	R208	R215	R221	R228
R1013	RU 35	AIO-PP8 · GH #2	R108	R115	R121	R128	R208	R215	R221	R228
R1113	RU 35	AIO-PP9 · IJ #1	R108	R115	R121	R128	R208	R215	R221	R228
R1213	RU 35	AIO-PP10 · IJ #2	R108	R115	R121	R128	R208	R215	R221	R228
R1313	RU 35	AIO-PP11 · KL #1	R108	R115	R121	R128	R208	R215	R221	R228
R1413 bms	RU 35	AIO-PP12 · KL #2	R108	R115	R121	R128	R208	R215	R221	R228
R1513	RU 39	AIO-PP1 · AB #1	R108	R115	R121	R128	R208	R215	R221	R228
R1613	RU 39	AIO-PP2 · AB #2	R108	R115	R121	R128	R208	R215	R221	R228
R1713	RU 39	AIO-PP3 · CD #1	R108	R115	R121	R128	R208	R215	R221	R228
R1813	RU 39	AIO-PP4 · CD #2	R108	R115	R121	R128	R208	R215	R221	R228
R1913	RU 39	AIO-PP5 · EF #1	R108	R115	R121	R128	R208	R215	R221	R228
R2013	RU 39	AIO-PP6 · EF #2	R108	R115	R121	R128	R208	R215	R221	R228
R2113	RU 39	AIO-PP7 · GH #1	R108	R115	R121	R128	R208	R215	R221	R228
R2213 bms	RU 39	AIO-PP8 · GH #2	R108	R115	R121	R128	R208	R215	R221	R228

AIO rack	DCN-PP panel	Lane	Col A	B	C	D	E	F	G	H
R2313	RU 39	AIO-PP9 · IJ #1	R108	R115	R121	R128	R208	R215	R221	R228
R2413	RU 39	AIO-PP10 · IJ #2	R108	R115	R121	R128	R208	R215	R221	R228
R2513	RU 39	AIO-PP11 · KL #1	R108	R115	R121	R128	R208	R215	R221	R228
R2613	RU 39	AIO-PP12 · KL #2	R108	R115	R121	R128	R208	R215	R221	R228

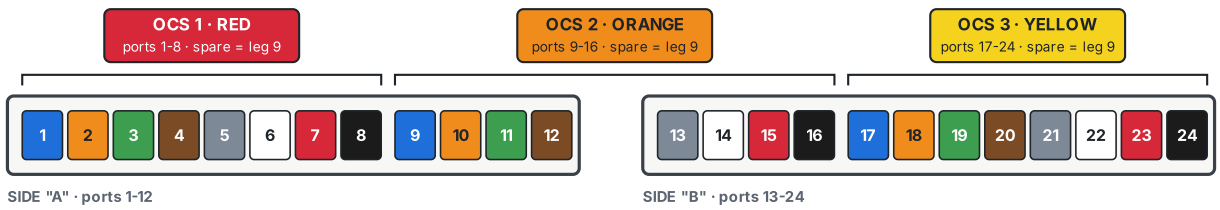
6. DCN-PP panels and spines (guide p.38-41)

- DCN-PP rack: 52 RU, eight 4RU panels (12 rows, columns A-L, 2 ports each). 2×2 shuffle modules, **P/N 760262482** (stop if 760262479 = 4×4).
- Rear: odd legs in the 1st column of the pair, even legs in the 2nd; legs go up; legs 19-24 spares; one lane per trunk; 12 AIO trunks per panel; two different AIO trunks per module.
- Front: each module's 4 ports (a1 a2 b3 b4, c5 c6 d7 d8, e9 e10 f11 f12 ...) → spines 1, 2, 3, 4 of the group; each row → next line-card port (cutsheet gives it).

DCN-PP	Racks	Spines
A	R107 / R108	1-4
B	R114 / R115	5-8
C	R120 / R121	9-12
D	R127 / R128	13-16
E	R207 / R208	17-20
F	R214 / R215	21-24

DCN-PP	Racks	Spines
G	R220 / R221	25-28
H	R227 / R228	29-32

7. OCS cables (guide p.15, 18-23)



OCS panel pp1.

Jacket	Cable	pp1 ports (TPU end)	Leg labels	Goes to
Red	OCS 1	1-8 (side A)	1-8	SPOCS rack 1
Orange	OCS 2	9-12 (side A) + 13-16 (side B)	9-16	SPOCS rack 2
Yellow	OCS 3	17-24 (side B)	17-24	SPOCS rack 3

18F Type B, 9 LC/APC duplex legs each end. Leg position colors 1-8: blue, orange, green, brown, slate, white, red, black; position 9 yellow = spare (fibers 17-18). In each group blue = lowest port, black = highest. Check reel part number and length before pulling.

Legs (position)	Length	SPOCS slot	Ports
1-2	1480 mm (~4 ft 10 in)	1 (top)	odd leg on **X**, even leg on **X+1** — same X on all 4 slots and all 3 SPOCS racks
3-4	1840 mm (~6 ft 0 in)	2	

Legs (position)	Length	SPOCS slot	Ports
5-6	2200 mm (~7 ft 3 in)	3	
7-8	2560 mm (~8 ft 5 in)	4 (bottom)	

- OCS shelf: 320 LC ports 001-320, pairs odd-over-even, count down a column of 20 then right; 145-160 not used. Left half 001-160 = TPU 1-72 (exit/dress left); right half 161-320 = TPU 73-144 (exit/dress right).
- Pull from the OCS rack to the TPU rack. Split point 12-18 in above the OCS rack. Tier 1 tray. OCS slack on top of the TPU rack. Spares: coil on top of the rack (drawings) or "left hanging" (reference cutsheet) — ask your lead.
- Labels: wrap-around on the bundle 1-2 in from the boot; flag on each leg 4 in from the boot.

8. Copper home runs (guide p.14, 26-27)

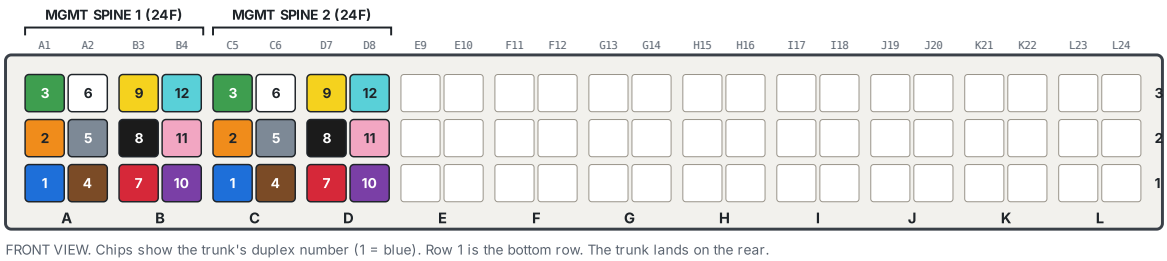
From	Port	EOR copper panel	Switch
TOR (side A)	ETH (Management1)	RU 24	MGMT leaf
TOR	CONSOLE	RU 30	console switch
TROSS / RMU1 (side A, RU 38)	GIG E 1	RU 22	TROSS leaf
RMU1	CONSOLE	RU 30	console switch
TROSS / RMU2 (side B, RU 38)	GIG E 1	RU 22	TROSS leaf

From	Port	EOR copper panel	Switch
RMU2	CONSOLE	RU 30	console switch

Home run: RJ45 plug at the TPU rack, straight to the jack the cutsheet gives (no panel at the TPU end). Patch cord jack N → switch port N. Flag label 4 in from the plug at both ends. Velcro, VCM fingers closest to the switch, nothing on the floor. Tier 2 at this site (p.14 says Tier 1 — ask).

9. Management trunks and SPOCS management (guide p.28-30)

Rack type	Trunk	Spine 1 (RU 39 rear)	Spine 2	Duplex assignments (same on both trunks)
AIO EOR	12F (6 duplex)	A1:A2, rows 1-3	C5:C6, rows 1-3	1 MGMT leaf · 2 TROSS leaf · 3-6 spares
AIO EOR w/BMS	24F (12 duplex)	A1:B4, rows 1-3	C5:D8, rows 1-3	1 MGMT leaf · 2 TROSS leaf · 3 FB0 BMS · 4-9 SPOCS mgmt · 10-12 spares



FRONT VIEW. Chips show the trunk's duplex number (1 = blue). Row 1 is the bottom row. The trunk lands on the rear.

RU 39 LC panel, 24F layout (front view, duplex numbers).

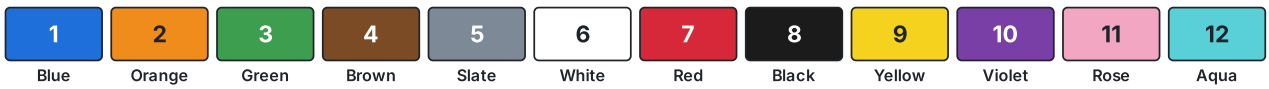
Pod	SPOCS racks	mgmt1 → (spot 11)	mgmt2 → (spot 13)
10	512, 612, 712	611	613

Pod	SPOCS racks	mgmt1 → (spot 11)	mgmt2 → (spot 13)
11	1312, 1412, 1512	1411	1413
12	2112, 2212, 2312	2211	2213

SPOCS uplinks: long 2F LC cords to the front of the w/BMS RU 39 panel, position from the cutsheet; each switch has 2 uplinks (one toward each spine). At R131/R204: trunk lands on a panel; breakout harness joins 4 LC legs into one MPO-8 at the spine port. Final patching before rack turn-up.

10. Labels and color codes (guide p.8)

- Machine printed only. Calibri Bold, 8 pt min on 2-in labels, 10 pt on larger labels.
- Line 1 = A device and port. Line 2 = Z device and port. Same text both ends, matching the cutsheet.
- Flag label (single connector, breakout leg, copper): 4 in from the boot, both sides printed, barcode at bottom. Wrap-around (trunks): 1-2 in from the boot at each end, 1 in wide minimum.
- Port lists: [1:4] = 1 through 4 · [1,3,5,7] = list · / separates line card / port / channel.
- Copper colors: orange OOB management · blue BMS · green FIS · black intra-rack (check site stock).



TIA-598 colors 1-12. Positions 13+ repeat — always read the number.

11. Dressing rules (DRS 01.00.01, 01.00.03,

01.00.03.01, 01.00.02 and guide p.7)

DRS 01.00.01 General Fiber Cable Dressing (revised 6/9/2026)

1. Dress cables per **ANSI/TIA-568, ANSI/TIA-606 and ANSI/TIA-942**.
2. **1/2-inch hook-and-loop (Velcro) only.** Nylon zip ties are not allowed.
3. **Bend radius:** follow the manufacturer's spec; minimum **20× diameter loaded** (under tension), **10× unloaded** (no tension). See DRS 01.00.07 for fiber bend radius.
4. Supports must not damage the cable. **No metal staples.**
5. **Support every 4-5 ft** on horizontal runs (TIA-568 recommendation).
6. **Rated cable:** plenum OFNP / CMP in plenum spaces; riser OFNR / CMR in vertical risers; LSZH where local code requires.

Cable diameter (example)	Loaded minimum radius (20×)	Unloaded minimum radius (10×)
0.20 in	4 in	2 in
0.25 in	5 in	2.5 in
0.30 in	6 in	3 in
0.40 in	8 in	4 in
0.50 in	10 in	5 in

Use your cable's real diameter. If the manufacturer's spec is larger, use the spec. A radius is half the width of the loop.

DRS 01.00.03 Cable Dressing Basket Trays (revised 6/9/2026)

1. Bundle neatly in the tray with **Velcro every 12-18 in**.
2. **Do not secure bundles to the tray** except at corners, exits and vertical runs: **3× at each corner, 3× at each exit**.
3. **1/2-inch Velcro** for all cable bundling.
4. **Waterfalls centered over the rack VCMs.**
5. Keep the **manufacturer's recommended bend radius** at all times (drawing: 20× cable diameter at a tray corner).

Never let cable drop over a tray edge unsupported. Waterfalls keep the bend radius at every tray exit.

DRS 01.00.03.01 Cable Dressing Vertical Pathways (revised 6/9/2026)

1. Velcro every 12-18 in, in vertical tray as well.
2. **Secure bundles to the tray when running vertically** (horizontal bundles float).
3. Vertical cable **secured to be self-supporting**: too much weight on horizontal transitions damages tray and cable.
4. **Waterfalls at all vertical transitions.**
5. **3/4-inch Velcro may secure cable to vertical tray** (bundling stays 1/2-inch).

Drawing: all bends greater than 20× cable diameter. Too little Velcro lets cable drop to the bottom.

DRS 01.00.02 Front of Rack Cable Dressing (revised 6/9/2026)

1. **Never cross devices you do not patch into.** Every device must be replaceable without unpatching another device's cables.
2. Exit the basket tray **directly above the rack edge or VCM**.
3. **Waterfalls** support every cable entering or leaving the tray (bend radius); each lines up with the rack VCM.
4. **Over-the-top waterfalls** at network racks (guide p.39, 41: DCN-PP and spine racks).
5. Bundle neatly with **1/2-inch Velcro**.
6. Use the waterfall **on the side of the rack you patch into**.

Tray accessories must fit trays made by Basor, WBT, Legrand or equivalent: 2 × 4 in (50 × 100 mm) openings, 5 mm wire.

VCM, slack and tiers (guide p.7; their DRS sheet has not been added yet)

9. VCM: exit between the fingers closest to the device; slack downstream; nothing touches the floor.
10. Trunk slack above the split point on top of the rack, never in the tray.
11. Tiers from the bottom: Tier 1 OCS · Tier 2 144F + CAT6 (this site).

Guide p.7 also allows 3/4-inch Velcro on vertical tray (DRS 01.00.03.01 agrees: for securing cable to vertical tray) and recommends 15× long-term bend radius (not on any DRS so far: open item 13).

Done-when checklists

TPU rack (guide p.16)

- ✓ 16 legs of the 144F trunk in TOR odd ports 1-31, in order (leg N in the Nth odd port).
- ✓ Legs 17 and 18 not plugged in at the TOR. Dressed and labeled. Not touching the floor.
- ✓ OCS cable 1 (red) in ports 1-8, cable 2 (orange) in 9-16, cable 3 (yellow) in 17-24.
- ✓ Three spare OCS legs handled the way your site decided. Labeled.
- ✓ 6 CAT6 cables: TOR ETH and console, RMU1 GIG E 1 and console, RMU2 GIG E 1 and console.
- ✓ Every fiber scoped and cleaned before it went in.
- ✓ Every cable flag-labeled at both ends, matching its cutsheet row.
- ✓ Nothing touched on the machine trays, batteries or rectifiers.

OCS cables (guide p.23)

- ✓ Right cable color to the right SPOCS rack: red 1, orange 2, yellow 3.
- ✓ Cable part number and length on the reel match the cutsheet.
- ✓ Legs 1-2 in slot 1, 3-4 in slot 2, 5-6 in slot 3, 7-8 in slot 4. Shortest legs at the top.
- ✓ Odd leg on port X, even leg on port X+1. Same X on all 4 slots and all 3 SPOCS racks.
- ✓ Spare leg (fibers 17-18) labeled and handled the way your site decided.
- ✓ Split point 12-18 inches above the OCS rack.
- ✓ TPU 1-72 dressed left. TPU 73-144 dressed right.
- ✓ Slack on top of the TPU rack, not in the tray. Nothing touches the floor.
- ✓ Every leg scoped and cleaned. Bundle and legs labeled at both ends.

EOR copper and management (guide p.31)

- ✓ Every copper home run terminated on the jack the cutsheet gives.
- ✓ A patch cord from each jack to the switch port with the same number.
- ✓ Management trunk to spine 1 in A1:A2 (12F) or A1:B4 (24F), rows 1-3.
- ✓ Management trunk to spine 2 in C5:C6 (12F) or C5:D8 (24F), rows 1-3.
- ✓ Switch cords in the front of the RU 39 panel: duplex 1 MGMT leaf, 2 TROSS leaf, 3 FB0 (w/BMS only).
- ✓ SPOCS management cords in the w/BMS panel (611/613, 1411/1413, 2211/2213): mgmt1 to spot 11, mgmt2 to spot 13.
- ✓ Far end patched at the management spine rack before turn-up.
- ✓ Trunk slack on top of the rack. Every cable labeled at both ends. Nothing touches the floor.

Large trunks (guide p.42)

- ✓ Each AIO front row = one TPU rack: TPU 1 on row 1, up to TPU 9 on row 9.
- ✓ 144F legs 1-16 left to right in columns A-H. Legs 17-18 in column I.
- ✓ Rows 10-12 and columns J-L left open on the front.
- ✓ One 192F trunk per column on the AIO rear: A = DCN-PP 1 … H = DCN-PP 8.
- ✓ 192F odd legs right, even legs left (rear view), legs going up. Legs 19-24 in rows 10-12.
- ✓ At the DCN-PP: 192F on the rear, in the trunk lane the cutsheet gives.
- ✓ Every 2×2 module holds trunks from 2 different AIO racks. Module P/N 760262482.
- ✓ Cords from the DCN-PP front to the spines patched per the cutsheet.
- ✓ Every leg scoped and cleaned. Trunk slack on top of the rack. Both ends labeled.

The guide's p.42 checklist says "192F odd legs left, even legs right"; its own p.37 drawing and notes say odd legs on the right port (rear view). The list above uses the rear-view wording from p.37 — see "Ask your lead" below.

Ask your lead: open items

#	Open item	Guide pages	What the course does
1	Revision mismatch: the cover says **Rev D - October 8, 2026**; every page footer says **Rev C, 10/8/2026**.	p.1 vs all footers	Notes both. Confirm you have the current revision.
2	CAT6 tray tier: Tier 2 at this site (recently changed) vs p.14 "Run copper in the Tier 1 tray".	p.7, 14, 16, 31, 47	Teaches Tier 2 (four places agree, one explains the change) and flags p.14.
3	OCS spare leg: drawings say coil on top of the rack; reference cutsheet says "left hanging".	p.15, 21, 22	Teaches "follow what your site decided" (as the guide does).
4	w/BMS management trunk size: one drawing says 12F; patching drawing and patch data say 24F.	p.30	Teaches 24F (the guide's choice); stop and ask if you see 12F at a w/BMS rack.
5	Shuffle module: patch data lists 2×2 P/N 760262482 (3 packs per panel); field reports mention a 4×4 (P/N 760262479).	p.40	Teaches the P/N check; stop if 760262479.
6	SPOCS uplinks: 24F table reserves 6 duplex positions (4-9) per trunk; the p.30 description (3 SPOCS racks × 1 switch × 1 uplink per spine) gives 3 per trunk.	p.29, 30	Teaches "use the cutsheet positions"; flags the count.
7	192F at the AIO rear: p.42 checklist says "odd legs left, even legs right"; p.37 says odd legs on the right port (rear view). Likely the same physical position described from different sides.	p.37, 42	Teaches the p.37 rear-view wording with the mirror explanation.
8	AIO elevations are not RU-exact (the drawing shows 2 blank units between the RU 39 LC panel and the 4RU MPO	p.26, 35	Uses the checked RU values (39, 32, 30, 24, 22); flags the rest.

#	Open item	Guide pages	What the course does
	panel at RU 32-35, which leaves RU 36-38; the guide names RU 36-37 as blank), show a 4th copper panel with no stated purpose, and do not show the console switch or the w/BMS FB0 switch / copper panel.		
9	Two port-code notations: cutsheet `r1-a1` (row, column, port) and AIO panel pre-label `A1-1` (column, port, row).	p.6, 36	Teaches both.
10	DCN-PP racks are numbered 1-8 (p.37) and lettered A-H (p.41, 44).	p.37, 41, 44	Teaches 1 = A ... 8 = H.
11	OCS "leg" numbering: position 1-8 (+9 spare) in the text vs printed labels 9-16 and 17-24 on cables 2 and 3.	p.15, 20, 21	Teaches both: label = pp1 port, color = position.
12	**Velcro width (answered by DRS 01.00.03.01).** Bundling: 1/2-inch everywhere (DRS 01.00.01 rule 2, 01.00.02 rule 5, 01.00.03 rule 3 "for all cable bundling"). Securing cable to vertical tray: 3/4-inch may be used (DRS 01.00.03.01 rule 5), matching guide p.7.	Guide p.7; DRS 01.00.01, 01.00.02, 01.00.03, 01.00.03.01	Teaches 1/2-inch to bundle and 3/4-inch only to secure to vertical tray. Confirm this reading.
13	**Bend radius 15× long-term.** Guide p.7 recommends 15× for long-term installed cable. DRS 01.00.01 gives 20× loaded / 10× unloaded (or the manufacturer spec) and points to DRS 01.00.07, which has not been supplied.	Guide p.7; DRS 01.00.01	Teaches the DRS minimums and shows 15× as the guide's recommendation. Ask for DRS 01.00.07.
14	**DRS numbering.** The guide lists DRS sections with two parts (01.00, 01.01, 02.00, 05.00, 10.01, 10.04, 10.08, 10.10; page drawings cite 01.02, 08.01 ...). The sheets use three parts (01.00.00 to	Guide p.1; DRS sheets	Cites DRS sheets by their full numbers. Ask your lead for the list of sheets per task.

#	Open item	Guide pages	What the course does
	01.00.03; 01.00.07 is referenced), and some add a fourth (01.00.03.01 builds on 01.00.03). Which sheets, and which current revisions, govern each of the five cable paths is not stated.		
15	**Support spacing wording.** The DRS 01.00.01 drawing says "TIA-568 recommends" support every 4-5 ft; the portal summary says "Support cable every 4-5 feet".	DRS 01.00.01	Teaches 4-5 ft as the DRS rule.
16	**Scope of DRS 01.00.01.** The drawing title says "General Fiber Cable Dressing"; the portal title says "General Cable Dressing", and rule 6 names copper ratings (CMP/CMR).	DRS 01.00.01	Applies it to every cable you dress; confirm whether copper has its own DRS.
17	**Having the DRS on the floor.** The DRS allows a printed copy or a digital copy on a device. Not stated: whether crews get portal logins or printed sets, and whether phones or tablets are allowed in the hall.	DRS 01.00.00	Teaches "get or print it before you go onto the floor". Ask your lead how your crew gets its copies.
18	**Which racks are "network racks"?** DRS 01.00.02 rule 4 requires over-the-top waterfalls for network racks but does not list them. The guide puts the note on the DCN-PP and spine rack pages (network rows 1-2).	DRS 01.00.02; guide p.39, 41	Teaches over-the-top at DCN-PP, spine and management spine racks in network rows 1-2. Ask about EOR (AIO) and SPOCS racks.
19	**"Should" on a DRS.** DRS 01.00.02 rule 6 says cables *should* use the waterfall on their side; the other rules say shall or must.	DRS 01.00.02; DRS 01.00.00	Teaches it as required (DRS 01.00.00: every DRS is a requirement). Confirm.
20	**Bend radius in tray pathways.** The DRS 01.00.03 drawing marks 20×	DRS 01.00.01, 01.00.03,	Teaches: in tray pathways (corners,

#	Open item	Guide pages	What the course does
	cable diameter at a tray corner, and the DRS 01.00.03.01 drawing says "all bends greater than 20× diameter of cable". DRS 01.00.01 gives 20× loaded and 10× unloaded, plus the manufacturer's spec and DRS 01.00.07.	01.00.03.01	vertical transitions) keep every bend above 20×, or the spec if larger. Confirm.

Details the guide does not give

Missing detail	Why it matters	What to provide or ask
Which physical rack is TPU 1-144 in each pod	Decides OCS port X and left/right dressing	Cutsheet rows (device names include rack number in pod, e.g. rack1)
OCS port X for each TPU rack; copper jack numbers; spine line-card ports	Exact port assignments	Cutsheet / BOM Builder extract
Copper termination standard (T568A/B), jack type, copper test/certification	Termination and verification	Site standard or spec
Fiber test requirements beyond scope-and-clean	Verification	Site test plan
FB0 BMS switch and its copper panel: RU, connected devices, who cables them; meaning of "FB0"	BMS connections	Rack integration spec / BMS drawings
Which spine racks are spines 1-32; which MGMT rack (R131/R204) is spine 1	Final patching	Cutsheet
Meaning of TROSS, RMU, FIS, SASL, Oxide, FU, CDU	Vocabulary only	Ask your lead (CDU is commonly Coolant Distribution Unit)

Missing detail	Why it matters	What to provide or ask
Ports 305-320 on the OCS shelf: the guide marks only 145-160 unused, but 144 TPU racks × 2 ports = 288 ports	Avoid assuming unused ports	Cutsheet

Glossary

144F trunk — 144-fiber trunk

A trunk cable with 144 fibers. It splits into 18 MPO-8 legs (18 × 8 = 144). 16 legs are used and 2 are spares. One runs from each TPU rack's TOR switch to its EOR (AIO) rack. (guide p.12, 13, 46)

192F trunk — 192-fiber trunk

A trunk with 192 fibers in 24 MPO-8 legs. Each AIO rack sends 8 of them from the back of its MPO panel, one to each DCN patch panel rack (columns A-H). (guide p.34, 37, 46)

2×2 shuffle module

A cassette at the DCN patch panel that mixes 2 MPO-8 inputs (from 2 different AIO trunks) into 2 MPO-8 outputs, so every output carries fibers from both inputs. Each module holds 2 shuffles. Large sites use P/N 760262482. If a module reads 760262479 (the 4×4), stop and ask. (guide p.38, 40)

A / Z — A end / Z end

The two ends of one cable. A is where it starts, Z is where it finishes. Every cutsheet row and every label is written A first, Z second. (guide p.6, 8)

AIO rack — All-In-One rack

The end-of-row rack. Same rack as the EOR. It holds the MPO 4RU panel (144F in front, 192F out the back), the copper panels, the management switches and the RU 39 LC panel. (guide p.25, 34, 46)

AIO-PP number — AIO patch panel number

On a DCN patch panel, the slot number a particular AIO rack's trunk uses (AIO-PP1 to AIO-PP12). One DCN-PP panel holds 12 AIO trunks. (guide p.39, 45)

ANSI/TIA — American National Standards Institute / Telecommunications Industry Association

TIA writes telecommunications standards; ANSI accredits them as American National Standards. DRS 01.00.01 says cable must be dressed per ANSI/TIA-568, ANSI/TIA-606 and ANSI/TIA-942. TIA-598 (fiber color code) is also a TIA standard. (DRS 01.00.00, 01.00.01; public (TIA))

APC — Angled Physical Contact

A fiber connector whose glass end face is polished at an 8° angle. APC connectors are usually green. Never plug an APC connector into a UPC (usually blue) connector: the cores do not line up and the end faces can be damaged. OCS legs use LC/APC. (guide p.20; public (Fluke Networks))

Basket tray — wire mesh cable tray

The overhead tray made of welded wire that cables run in above the racks. In it, bundle with 1/2-inch Velcro every 12-18 inches and secure to the tray only at corners, exits and vertical runs, 3 times at each corner and exit (DRS 01.00.03). Tray accessories such as waterfalls must fit trays made by Basor, WBT, Legrand or equivalent, with 2 × 4 inch (50 × 100 mm) openings and 5 mm wire (DRS 01.00.02).

(DRS 01.00.02, 01.00.03; guide p.7)

Bend radius

How tightly a cable is allowed to curve, measured from the center of the curve out to the cable. DRS 01.00.01 rule 3: follow the manufacturer's spec, with at least 20 × the cable diameter when loaded (under tension, such as while pulling) and 10 × when unloaded. DRS 01.00.03 rule 5: keep the manufacturer's recommended bend radius at all times; its drawing marks 20 × at a tray corner, and the DRS 01.00.03.01 drawing says all bends greater than 20 ×. DRS 01.00.01 points to DRS 01.00.07 for fiber detail. The guide (p.7) adds 15 × as the long-term recommendation. (DRS 01.00.01, 01.00.03, 01.00.03.01; guide p.7)

BICSI — Building Industry Consulting Service International

A professional association for the ICT industry. It publishes design and installation standards and runs training and credentials. DRS documents reference BICSI. (DRS 01.00.00 (named); public (BICSI, ANSI))

BMS — Building Management System

A control system that monitors and manages a building's mechanical and electrical systems, such as power, cooling, water detection and fire safety. In this hall, six EOR racks are the "w/BMS" type and carry the BMS network switch (FB0 BMS) and its uplink. (guide p.25-26, 29; public (TechTarget))

BOM Builder — Bill of Materials Builder

The tool that lists every part and port for a site. The guide says exact OCS ports come from the cutsheet or BOM Builder. (guide p.21)

Boot

The rubber cover behind a connector. Label distances are measured from the boot. (guide p.8)

Breakout

A cable end where one connector splits into several smaller ones, such as one MPO plug splitting into several LC legs. (guide p.46)

CAT6 — Category 6 copper network cable

Copper network (Ethernet) cable with an RJ45 plug. Each TPU rack sends 6 CAT6 home runs to its EOR rack. (guide p.13, 46)

CDFP cable — a 400 Gb/s pluggable form factor (CD = 400 in Roman numerals)

High-speed cables used inside the TPU rack. They arrive pre-installed. You do not touch them. (guide p.4 (named only); general)

CDU — Coolant Distribution Unit (not defined in the guide)

The CDU blocks at the ends of each compute row on the hall map. In liquid-cooled data centers a CDU manages the flow, temperature and pressure of coolant to the racks. Not part of your cable work. (guide p.5 (map label only); public (Supermicro, Hot Chips 2025))

Column (panel)

A lettered position across a panel, A to L. On the MPO and LC panels each column has 2 ports. (guide p.35, 46)

Column pair

Two side-by-side columns on a DCN patch panel (AB, CD, EF, GH, IJ, KL) that hold one 2×2 module. (guide p.39)

Confluence (Fluidstack portal)

The authoritative source for all current DRS documents. Log in at fluidstack.atlassian.net, open the LVC Deployment space, then DEPLOYMENT REQUIREMENTS SHEETS. It needs a login and a connection. If you cannot get in, contact your Fluidstack PM or site supervisor. (DRS 01.00.00)

Console port

A port that gives a direct, text-only connection to a device for setup and recovery. The TOR and both TROSS/RMU units each send a console run to the EOR console panel (RU 30). (guide p.13-14; general (WTI))

Cutsheet

The spreadsheet that lists every cable and every stop it makes. Your site's cutsheet is the final word on any port. (guide p.1, 6)

DAC cable — Direct Attach Copper

Short copper cables inside the TPU rack. Pre-installed. Not yours. (guide p.4 (named only); general)

DCN — Data Center Network

The main data network. Its big switches are the DCN spines in network rows 1 and 2. (guide p.4, 46)

DCN-PP — DCN Patch Panel rack

A passive rack in the network rows where the 192F trunks end. It holds eight 4RU panels with 2×2 shuffle modules. Pairs: A (R107/R108) to H (R227/R228). The guide also numbers them DCN-PP 1-8 (1 = A ... 8 = H). (guide p.34, 38, 41, 44)

DH — Data Hall

One room of racks. This course covers Data Hall 4 (DH4). (guide p.5)

Downstream

The side of a connection toward the end device. Slack goes on the downstream side. (guide p.7)

DRS — Deployment Requirement Sheet

Fluidstack's required technical standard for how one on-site task must be done. Requirements, not guidelines. Each DRS has a number (such as 01.00.01), a title, a revision date and an approver, and references industry standards (ANSI/TIA, BICSI, NEC, NEMA). The current version is in Fluidstack Confluence. Have the applicable DRS on hand (printed or on a device) when you do the work; if you do not, stop work and get it. The guide was built from DRS sections 01.00, 01.01, 02.00, 05.00, 10.01, 10.04, 10.08 and 10.10. (DRS 01.00.00; guide p.1)

Duplex

One LC leg with 2 fibers (usually one to send and one to receive). A 12F trunk has 6 duplex legs; a 24F trunk has 12. (guide p.21, 28, 46)

Elevation

A front view of a rack showing every rack unit. Elevations are subject to change; the rack integration spec has the final view. (guide p.26, 35)

EOR rack — End Of Row rack

The rack at spot 11 or 13 of a compute row. Same rack as the AIO. Each EOR serves the 9 TPU racks on its side of the middle aisle. (guide p.11, 25, 46)

F (as in 144F) — fibers

The number in front of F is a fiber count. 144F = 144 fibers. 2F = 2 fibers (one LC duplex cord). (guide p.4, 46)

FB0 BMS switch — (FB0 is not expanded in the guide)

A switch for building systems, found only in AIO EOR w/BMS racks. Its uplink rides duplex 3 of the 24F management trunks. (guide p.26, 29, 46)

FIS (green copper) — (not defined in the guide)

The guide lists green copper as "FIS" without explaining it. Ask your lead if you see it. (guide p.8)

Flag label

A self-laminating label that sticks out from the cable like a flag. Used on single connectors, breakout legs and copper runs. Placed 4 inches from the boot, printed on both sides, barcode at the bottom. (guide p.8)

Fluidstack

The company whose data center deployments these DRS documents govern. Fluidstack writes and approves every DRS, and its DRS rules apply to every contractor and every site on its deployments. (DRS 01.00.00)

GIG E 1 — Gigabit Ethernet port 1

The network port on each TROSS / RMU that you cable to the EOR TROSS leaf panel (RU 22). (guide p.14)

Home run

One copper cable straight from a device in the TPU rack to a jack in the EOR rack, with no panel at the TPU end. (guide p.14, 27)

ICT — Information and Communications Technology

Networks, cabling and computing equipment. Fluidstack ICT leadership reviews and approves every DRS. (DRS 01.00.00 (named); general)

Jack

The socket on a copper panel. Each home run is terminated on a jack; a short patch cord connects the jack to the switch port with the same number. (guide p.14, 27)

Jacket

The outer cover of a cable. On OCS cables the jacket color is the cable's ID: red = OCS 1, orange = OCS 2, yellow = OCS 3. (guide p.20)

Lane / trunk lane

Two meanings in the guide: (1) one of the signals inside an optic port; (2) the ports one trunk uses inside a DCN-PP column pair (lane 1 or lane 2). (guide p.39, 46)

LC connector — LC (a small push-pull fiber connector)

A small push-pull fiber plug. LC duplex = 2 fibers side by side. OCS legs use the angled LC/APC type. (guide p.20, 46)

Leg

One connector at the end of a trunk or bundle. The 144F trunk has 18 legs, the 192F has 24, an OCS cable has 9 at each end. (guide p.6, 13, 46)

Line card

A slot in a spine switch that holds ports. Each row of a DCN-PP panel lands on the next line-card port of its 4 spines; the cutsheet gives the exact port. (guide p.41)

LSZH — Low Smoke Zero Halogen

Cable jacket material that gives off little smoke and no halogen gases when it burns. DRS 01.00.01: use LSZH where local code requires it. (DRS 01.00.01; public (cable manufacturers))

MGMT — Management

The small network used to control and monitor equipment. Also called OOB (out-of-band). (guide p.46-47)

MGMT leaf / TROSS leaf

Pre-installed switches in the AIO rack. The MGMT leaf takes the TOR management runs (RU 24 panel). The TROSS leaf takes the TROSS / RMU GIG E 1 runs (RU 22 panel). (guide p.26, 46)

MGMT spine — Management spine

A switch that gathers all management traffic. The management spine racks are R131 and R204. (guide p.25, 30)

MPO / MPO-8 — Multi-fiber Push-On (8-fiber)

A rectangular push-on plug that holds many fibers. MPO-8 holds 8. Every 144F and 192F trunk leg is an MPO-8. (guide p.13, 40, 46)

NEC — National Electrical Code (NFPA 70)

The U.S. electrical safety code, published by NFPA. Its Article 770 (optical fiber) and Article 800 (communications) set the fire ratings on cable jackets, such as OFNP plenum and OFNR riser. DRS documents reference the NEC. (DRS 01.00.00 (named); public (NFPA))

NEMA — National Electrical Manufacturers Association

A U.S. association that publishes standards for electrical products such as enclosures, wire and cable, and cable tray. DRS documents reference NEMA. (DRS 01.00.00 (named); public (NEMA))

OCS — Optical Circuit Switch

A shelf that steers light between fibers. Public research from Google describes these switches using tiny movable mirrors (MEMS), so racks can be reconnected in new shapes without anyone unplugging cables. In this guide "OCS" also names the 3 OCS cables and the OCS panel on the TPU rack. (guide p.18, 46; public (Mission Apollo paper))

OCS bundle

The guide's other name for an OCS cable: the 18-fiber cable with 9 LC legs at each end. (guide p.46)

Odd port

A port with an odd number: 1, 3, 5 ... On the TOR switch the odd ports are the top row and are the ports you use for the 144F trunk (1, 3, 5 ... 31). (guide p.13, 46)

OOB — Out-Of-Band

The management network: a separate path to reach and control equipment, apart from the main data network. Orange copper = OOB management at this site (check what your site stocks). (guide p.8, 47; general (WTI))

Outage

When a network or system stops working. Following the DRS lowers the risk of outages. (DRS 01.00.00; general)

Over-the-top waterfall

A waterfall that carries cable up and over the top edge of the basket tray. Required at network racks (DRS 01.00.02 rule 4). In this guide the note appears on the DCN patch panel and spine rack pages (p.39, 41). (DRS 01.00.02; guide p.39, 41)

Oxide, SASL, FU racks — (not defined in the guide)

Rack types shown in network rows 1-2 on the map. The guide does not explain them and you do not install anything in them. (guide p.5, 44)

Patch cord

A short cable with a plug on each end, used to connect a panel to a switch. (guide p.27, 47)

Plenum / riser / LSZH — OFNP / CMP (plenum), OFNR / CMR (riser), Low Smoke Zero Halogen

Fire ratings printed on the cable jacket. DRS 01.00.01 rule 6: plenum-rated (OFNP fiber / CMP copper) in plenum spaces, riser-rated (OFNR / CMR) in vertical risers, LSZH where local code requires. A plenum is a space that is part of the air-handling system (NEC Article 100). The ratings come from the NEC: Article 770 for optical fiber (OFN...), Article 800 for communications copper (CM...). A higher rating can replace a lower one (plenum where riser is asked for), never the other way round. (DRS 01.00.01; guide p.7; public (NFPA 70 / NEC))

Plenum space

A space that is part of the building's air-handling system, such as some spaces above ceilings or below raised floors (NEC Article 100). Only plenum-rated cable (OFNP / CMP) may run there (DRS 01.00.01 rule 6). (DRS 01.00.01; public (NFPA 70 / NEC))

PM — Project Manager

Your assigned Fluidstack project manager. With your site supervisor, the backup way to get a DRS: they can give you the document directly or restore your portal access. (DRS 01.00.00)

Pod

A group of 144 TPU racks that share 3 SPOCS racks. Data Hall 4 has pods 10, 11 and 12. (guide p.5, 33, 47)

Port X

Your TPU rack's OCS port number at the SPOCS racks, from the cutsheet. The odd leg goes on X and the even leg on X+1, on every slot of all 3 SPOCS racks. (guide p.21, 47)

PP — Patch Panel

A panel of ports where cables land so they can be connected onward. (guide p.26, 47)

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pp1 — the OCS panel on the TPU rack

The cutsheet's name for the TPU rack's OCS panel. It has ports 1-24: the drawings show 1-12 on side A and 13-24 on side B, but the cutsheet lists them all as one panel. (guide p.12, 15)

Rear view

A panel seen from the back of the rack. Column A is on the right, so you work right to left. (guide p.37, 47)

Rectifier

A power unit that converts incoming AC power to the DC power the rack uses. Pre-installed. Not yours. (guide p.12 (named); general; public (Hot Chips 2025))

Revision (DRS)

A new version of a DRS. When a standard changes, a new revision is issued and you are notified through the portal. Always use the current version: check the revision date on the cover page (DRS 01.00.01 says REVISED 6/9/2026). (DRS 01.00.00, 01.00.01)

Riser

A vertical pathway that runs between floors. Riser-rated cable (OFNR / CMR) is required there (DRS 01.00.01 rule 6). (DRS 01.00.01; public (NFPA 70 / NEC))

RJ45 — Registered Jack 45 (the common name for the 8-pin modular plug)

The square plug on a copper network cable. The TPU end of each home run is an RJ45 plug. (guide p.14)

RMU — (cutsheet name, not expanded in the guide)

The cutsheet's name for the TROSS units: rmu1 on side A and rmu2 on side B. (guide p.12)

Row

Two meanings: a row of racks in the hall (row 3, row 4 ...), or a numbered position up a panel (row 1 at the bottom). (guide p.47)

RU — Rack Unit

One 1.75-inch slot in a rack. RU 1 is at the bottom. A "4RU" panel is 4 rack units tall. (guide p.6, 47)

Scope

The handheld camera used to look at a fiber end face. Scope and clean every leg before it goes in. (guide p.13, 47)

Self-supporting

Secured along its length so each section of a vertical run carries its own weight, instead of hanging from the top or piling onto the bend at the bottom (DRS 01.00.03.01 rule 3). (DRS 01.00.03.01)

Shall / must / should (DRS wording)

Shall and must mean required. In standards writing, should usually means recommended. DRS 01.00.00 says every DRS is a requirement, so treat a should on a DRS (such as DRS 01.00.02 rule 6) as required, and ask your lead if you think it cannot be met. (DRS 01.00.00, 01.00.02; general (standards wording))

Slot

One of the 4 OCS shelf positions in a SPOCS rack. Slot 1 is the top shelf. (guide p.19, 47)

Spare

A leg that is run and labeled but not plugged in. A cutsheet row that says SPARE is a leg you land and label but do not plug into a device. (guide p.6, 47)

Spine

A big DCN switch in network rows 1-2. Every TPU rack reaches every spine. There are 32 DCN spines in the groups listed on guide p.41. (guide p.41, 47)

Split point — also called the furcation point

Where a trunk or bundle breaks into legs. Trunk slack above the split point is stored on top of the rack. For OCS bundles the split point is 12-18 inches above the OCS rack. (guide p.22, 47)

SPOCS — Super Pod OCS rack

A rack that holds 4 OCS shelves. Each pod has 3: red = SPOCS 1, orange = SPOCS 2, yellow = SPOCS 3. They sit at spot 12 (pod 10: 512, 612, 712; pod 11: 1312, 1412, 1512; pod 12: 2112, 2212, 2312). (guide p.18-19, 47)

Spot

A rack's place in its row. Rack 512 is row 5, spot 12. (guide p.5, 47)

TIA-568 — Commercial Building Telecommunications Cabling Standards

The main TIA standard for building telecom cabling: how cable is installed and supported. DRS 01.00.01 names it for dressing, and its drawing says TIA-568 recommends support every 4-5 ft on horizontal runs. (DRS 01.00.01; public (TIA))

TIA-598 — TIA-598 fiber color code standard

The standard fiber color order: 1 blue, 2 orange, 3 green, 4 brown, 5 slate, 6 white, 7 red, 8 black, 9 yellow, 10 violet, 11 rose, 12 aqua. Positions 13 and up repeat the colors, so always read the number on the label. (guide p.8, 13, 47; public (FOA))

TIA-606 — Administration standard for telecommunications infrastructure

The TIA standard for administration: identifiers, labels and records, so every cable and port can be traced. DRS 01.00.01 names it for dressing. Your site's label format comes from the guide (p.8) and the cutsheet. (DRS 01.00.01; public (TIA))

TIA-942 — Telecommunications Infrastructure Standard for Data Centers

The TIA standard for the infrastructure of data centers and computer rooms, including cabling and pathways. DRS 01.00.01 names it for dressing. (DRS 01.00.01; public (TIA))

Tier

A tray level, counted from the bottom up. Tier 1: OCS cables. Tier 2: 144F trunks and, at this site, CAT6 (guide p.14 still says Tier 1 for copper: ask your lead). (guide p.7, 14, 47)

TOR switch — Top Of Rack switch

The TPU rack's door to the data network. An Arista 7060 on side A. The 144F trunk plugs into its odd ports (top row); it also sends 2 copper runs (ETH and console). (guide p.12-13, 47)

TPU — Tensor Processing Unit

A computer chip built for AI work. In this guide "TPU rack" (or just "TPU") means a rack of these AI computers. The drawings call it an Ironwood TPU rack. (guide p.11-12, 47; public (Google Cloud))

TROSS / RMU — (TROSS is not expanded in the guide)

The TPU rack's management box, one per side at RU 38. The drawings call it TROSS; the cutsheet calls it rmu1 (side A) and rmu2 (side B). Each sends 2 CAT6 runs: GIG E 1 and CONSOLE. (guide p.12, 14, 47)

Trunk

A thick multi-fiber cable that splits into legs at each end. Gets a wrap-around label 1-2 inches from the boot at each end. (guide p.6, 8, 47)

Turn-up

When a rack is powered and brought online. Final patching at the management spine rack must be done before turn-up. (guide p.30, 47)

Type B — Type B polarity

A cable that crosses send and receive, so one end's transmit fiber meets the other end's receive port. OCS cables are Type B. The cutsheet already accounts for it; you do not swap anything. (guide p.20, 47; public (SENKO))

Uplink

A connection from a switch up to a bigger switch. (guide p.30)

VCM — Vertical Cable Manager

The channel on the side of a rack. Cable leaves the tray directly above the rack edge or the VCM (DRS 01.00.02 rule 2), over a waterfall centered over the VCM (DRS 01.00.03 rule 4). Cables exit between the fingers closest to the device they patch (guide p.7). (DRS 01.00.02, 01.00.03; guide p.7, 47)

Velcro — hook-and-loop strap

The only tie allowed. Bundle cables with 1/2-inch hook-and-loop everywhere (DRS 01.00.01 rule 2; DRS 01.00.02 rule 5; DRS 01.00.03 rule 3: for all cable bundling), 12-18 inches apart in tray (DRS 01.00.03 rule 1; DRS 01.00.03.01 rule 1). 3/4-inch Velcro may be used to secure cable to vertical tray (DRS 01.00.03.01 rule 5; guide p.7). Nylon zip ties are not allowed. (DRS 01.00.01, 01.00.02, 01.00.03, 01.00.03.01; guide p.7)

Vertical pathway — vertical tray or ladder

A tray or ladder that carries cable up or down between levels. DRS 01.00.03.01: bundle with Velcro every 12-18 inches, secure the bundles to the tray so the cable is self-supporting, use waterfalls at all vertical transitions; 3/4-inch Velcro may be used to secure cable to vertical tray. Too little Velcro lets the cable slide to the bottom and damage the cable and tray. (DRS 01.00.03.01)

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w/BMS — with Building Management System

An EOR rack type with extra parts: the FB0 BMS switch and a copper panel. Six in this hall: 611, 613, 1411, 1413, 2211, 2213. Their management trunks are 24F. (guide p.25, 29, 47)

Waterfall

A curved guide on the edge of the tray that supports cable where it enters or leaves the tray, so the bend stays wide. Never let cable drop over a tray edge

unsupported. Waterfalls are centered over the rack VCMs (DRS 01.00.03 rule 4), on the side of the rack the cables patch into (DRS 01.00.02 rules 3, 6), and required at all vertical transitions (DRS 01.00.03.01 rule 4). Network racks use over-the-top waterfalls (DRS 01.00.02 rule 4). (DRS 01.00.02, 01.00.03, 01.00.03.01; guide p.7, 47)

Wrap-around label
A label that circles the cable. Used on trunks, 1-2 inches from the boot of each end connector, at least 1 inch wide. (guide p.8, 47)

Sources & coverage

Project sources

Fluidstack DRS (Deployment Requirement Sheets), Fluidstack Internal Use Only, supplied one section at a time:

- **DRS 01.00.00 — introduction** (portal page): what a DRS is, what it contains, why it exists, approval (Tom Marsh, Infrastructure Deployment Manager, 6/9/2026), revisions, where to get it, and the on-site requirement. Used in Lesson 0.
- **DRS 01.00.01 — General Fiber Cable Dressing** (drawing, REVISED 6/9/2026, approved by Tom Marsh) and its portal summary "General Cable Dressing". Used in Lessons 0, 3 and 5.
- **DRS 01.00.02 — Front of Rack Cable Dressing** (drawing, REVISED 6/9/2026, approved by Tom Marsh) and its portal summary. Used in Lessons 5B and 9.
- **DRS 01.00.03 — Cable Dressing Basket Trays** (drawing, REVISED 6/9/2026, approved by Tom Marsh) and its portal summary "Basket Tray Cable Dressing". Used in Lessons 5B, 11 and 12.
- **DRS 01.00.03.01 — Cable Dressing Vertical Pathways** (drawing, REVISED 6/9/2026, approved by Tom Marsh) and its portal summary "Vertical Pathway Dressing". Used in Lessons 5, 5B and 11.

The DRS sheets page lists every sheet in plain words with where it is taught.

Where the DRS and the guide differ, the course teaches the DRS and lists the difference under Open items.

Large Site Cable Installation Guide, data hall 4 (pods 10, 11, 12), D. Moore — cover Rev D (October 8, 2026), page footers Rev C (10/8/2026), 47 pages. Every project-specific instruction, port mapping, color sequence and quiz answer in this course comes from this guide. Page numbers are cited throughout.

The guide says it was built from the DRS (sections 01.00, 01.01, 02.00, 05.00, 10.01, 10.04, 10.08, 10.10) and the BUF102 patch panel matrix v0.97 (ICT cutsheet v0.97). Apart from DRS 01.00.00 to 01.00.03.01, those documents have not been supplied, so the guide's "checked against the patch data" statements are reported as the guide states them.

Public sources used for background only

These explain concepts (what a TPU or an OCS is, why APC connectors must not mix with UPC, and so on). They never replace the guide's wiring instructions.

- Google Cloud — TPU7x (Ironwood) documentation: seventh-generation TPU; 9,216 chips per pod; 64-chip 4×4×4 cubes.
- Google — 3 things to know about Ironwood: superpods of up to 9,216 chips.
- ServeTheHome — Ironwood at Hot Chips 2025: 16 trays × 4 TPUs = 64 per rack; liquid-cooled; copper inside the rack, OCS between racks.
- Hot Chips 2025 tutorial (Google data center racks): OCS rack with OCS1-OCS4, CDU (Coolant Distribution Unit), AC/DC rectifiers, battery backup.
- Mission Apollo: Landing Optical Circuit Switching at Datacenter Scale (arXiv): 3D MEMS-mirror optical circuit switches.
- Fiber Optic Association — color codes: TIA-598 order; APC = green connector; singlemode = yellow jacket.
- Fluke Networks — What is an APC connector: 8° angle; never mate APC with UPC.
- VIAVI — What is fiber inspection: contamination is the #1 reason to troubleshoot optical networks; inspect both sides.
- SENKO — MPO connector polarity (TIA-568): transmit must reach receive; Type A/B/C.

- TechTarget — Building management system: what a BMS monitors and controls.
- Supermicro — Coolant distribution unit.
- WTI — Out-of-band management in the data center: OOB vs in-band; console ports.

Standards named in the DRS (background only)

These explain what each standard covers. The DRS turns them into the rules you follow; on site, follow the DRS.

- TIA — TIA-568: Commercial Building Telecommunications Cabling Standards.
- TIA — TIA-942: "specifies the requirements for infrastructure of data centers and computer rooms".
- Electrical Contractor — TIA redefines telecommunications administration: TIA-606 defines how identifiers, labels and records work together.
- Cabling Installation & Maintenance — Boiling down the 606-A labeling standard: identifiers and labels; handwritten labels are not acceptable.
- EC&M — Plenum datacom wiring: NEC Article 100 definition of a plenum; higher-rated cable may substitute for lower.
- Open Exam Prep — Electrical protection and the NEC: NEC Articles 770 and 800; CMP for plenums, CMR for floor-to-floor risers; OFNP above OFNR; never substitute a lower rating.
- Low smoke zero halogen (Wikipedia): LSZH jackets give off limited smoke and no halogen when heated.
- ANSI Standards Portal — BICSI: professional association for the ICT profession; ANSI-accredited standards developer; training and credentials.
- NEMA — Standards by product: product standards including enclosures, cable tray, wiring devices, and wire and cable.

Coverage map: guide → lesson → visual → practice → video

Guide page(s)	Topic	Lesson / section	Visual	Practice	Video
1	Scope, revision, guide sources	L1 s1; L15 s3	overview	—	V01, V15
4	Your job: five cable paths; do not touch	L1 s2-s5	overview	a1-explore, a1-match, a1-mistake	V01 ch.2-4
5, 11, 18, 25, 33	Hall map, location codes, row pattern, pods	L2 s1-s5	hallMap, locationCode	a2-pick, a2-decode	V02
6	Device names, port codes, cutsheet row, SPARE	L4 s1-s3, s5	cutsheet	a4-wrong	V04 ch.1-3
DRS 01.00.00	What a DRS is, contents, standards, approval, revisions, where to get it, on-site rule	L0 all	drsStack, drsAccess, drsSheet, standardsMap	a0-order, a0-explore, a0-scen, a0-match	V00
DRS 01.00.01	General cable dressing: standards, Velcro, bend radius, supports, spacing, rated cable	L5 s1-s4, s6-s10; L3 s6	dressRules, standardsMap, bend, cableRatings	a5-explore, a5-bend, a5-match, a5-spot	V05; V03 ch.7
DRS 01.00.02	Front of rack dressing: no crossing, tray exit, waterfalls, over-the-top at network racks, Velcro, side, tray accessories	L5B s1, s4, s6-s9; L9 s3, s5	frontRack	a5-front, a5b-match, a5b-spot	V05B ch.3
DRS 01.00.03	Basket tray dressing: Velcro every 12-18 in, secure 3× at corners	L5B s1-s2, s6-s9; L11 s6; L12 s5	trayPlan	a5b-tray, a5b-	V05B ch.2; V11

Guide page(s)	Topic	Lesson / section	Visual	Practice	Video
	and exits only, 1/2-in Velcro, waterfalls centered over VCMs, bend radius, no unsupported drops			match, a5b-spot	
DRS 01.00.03.01	Vertical pathways: Velcro 12-18 in, secured, self-supporting, waterfalls at all vertical transitions, 3/4-in Velcro to secure, bends over 20×	L5B s1, s3, s6, s8-s9; L5 s3; L11 s6	vertPath	a5b-vert, a5b-match, a5b-spot	V05B ch.3
7	Cable dressing rules, tiers, bend radius	L5 s3, s5-s6; L5B s5; L3 s6	tray, bend	a5-order, a5-spot	V05; V05B; V03 ch.7
8	Labels, TIA-598, copper colors	L4 s4; L3 s4-s5	labels, colorStrip	a4-labels, a3-order, a3-scen	V04 ch.4; V03 ch.4-5
9	Guide set and install order	Home; L1 s3	—	—	V01
12	TPU rack: sides A/B, units, what is yours	L6 s1-s4	tpuRack	a6-explore, a6-pick	V06
13	144F trunk at the TOR: odd ports 1-31, legs, spares, colors	L7 all	trunk144, torSwitch	a7-pick, a7-mistake	V07
14	TROSS / RMU copper; all 6 copper cables	L12 s2-s6	rmuFace, mgmtFlow	a12-match, a12-scen	V12

Guide page(s)	Topic	Lesson / section	Visual	Practice	Video
15	OCS panel pp1, port groups, spare conflict	L10 s4-s8	ocsPanel	a10-pick, a10-mistake	V10 ch.5-7
16, 23, 31, 42	Done-when checklists	L15 s2; Tables	—	a15-scen	V15 ch.4
18-19	SPOCS racks, slots, OCS shelf 320 ports	L10 s2; L11 s1-s2	spocsRack, ocsShelf	a11-scen	V10 ch.2; V11 ch.1-2
20	OCS cable: 18F Type B, 9 LC/APC legs, jacket colors, labels	L10 s3-s4	ocsCable	a10-pick	V10 ch.3-4
21	Patching one OCS cable: lengths, slots, X / X+1	L11 s3-s5	spocsRack (legs), ocsShelf	a11-match, a11-scen	V11 ch.3-5
22	Dressing OCS: split point, left/right, Tier 1, slack, spares	L11 s6	ocsShelf	a11-scen	V11 ch.6
26, 35	AIO EOR rack elevation and panel RUs	L8 s1-s2; L12 s4	aioRack	—	V08 ch.1
27	Landing copper runs: jack = port, patch cords	L12 s4	copperPanel	a12-scen	V12 ch.4
28	Management trunks 12F	L13 s1-s4	lcPanel (12F)	a13-scen	V13 ch.1-2
29	Management trunks 24F (w/BMS), duplex assignments	L13 s2-s4; L14 s2	lcPanel (24F)	a13-pick	V13 ch.2; V14 ch.2
30	SPOCS management uplinks; 12F/24F correction	L13 s5	spocsMgmt	a13-scen	V13 ch.4

Guide page(s)	Topic	Lesson / section	Visual	Practice	Video
25-26, 29 (BMS)	BMS: w/BMS racks, FB0 switch, uplink path	L14 all	bmsFlow, hallMap	a14-trace	V14
34	Large trunk path at a glance	L8; L9; L15 s1	largePath	a15-trace	V15 ch.2
36	AIO front: 144F, one TPU per row	L8 s3; L7 s6	aioFront	a8-front	V08 ch.2; V07 ch.7
37	AIO back: 192F, mirror rule, line-up	L8 s4-s5	aioRear, aioFront	a8-scen	V08 ch.3-4
38-39	DCN-PP rack, landing 192F, lanes	L9 s1-s3	dcnppRear	a9-lookup	V09 ch.1-3
40	2×2 shuffle module, P/N check	L9 s4	shuffle	a9-lookup	V09 ch.4
41	DCN-PP front to spines, spine groups	L9 s5-s6	dcnppFront	a9-lookup	V09 ch.5-6
43-45	Trunk Landing Sheet (all 48 AIO racks)	L8 s6; L9 s2; Tables §5	networkRows	a8-scen, a9-lookup	V08 ch.5; V09 ch.2
46-47	Glossary	Glossary page; term pop-ups throughout	—	—	Term cards in every video

Mandatory focus topics

Topic	Lessons	Videos	Practice
DRS — what it is, where to get it, on-site rule; dressing per DRS 01.00.01,	0, 5, 5B	V00, V05, V05B	Order the steps; explore a DRS sheet; which source

Topic	Lessons	Videos	Practice
01.00.02, 01.00.03 and 01.00.03.01 and the standards they name			answers it; bend radius; which rule says so
TPU — what it is, the rack, what is yours	6 (and 1)	V06	Explore the rack; click the parts
144-fiber arrangement, odd ports 1-31, both ends	7, 8	V07, V08	Plug legs into TOR ports; find the wrong leg; AIO front ports
OCS — purpose, cables, colors, ports, both ends	10, 11	V10, V11	pp1 port picking; find the swapped legs; slots and X/X+1
Copper / Ethernet	12	V12	Match runs to panels; spot problems
BMS	14 (and 13)	V14	Trace the FB0 BMS uplink

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